S^n, ~~$S^1 \times S^1$~~, $\mathbb{RP}^n$,

$(n \geq 1)$
$\Downarrow$
$\pi_1 = 0$.

2-dim mfd

$L(p,q)$

$S^1 \times S^1 / e^{2\pi i p}, e^{2\pi i q}$

$M \# N$. $\wedge$ $\vee$

$[0,1]^2$

$\mathbb{Z} \times \mathbb{Z}$ torus.

Klein.

Klein. free product.

$ababab b$.

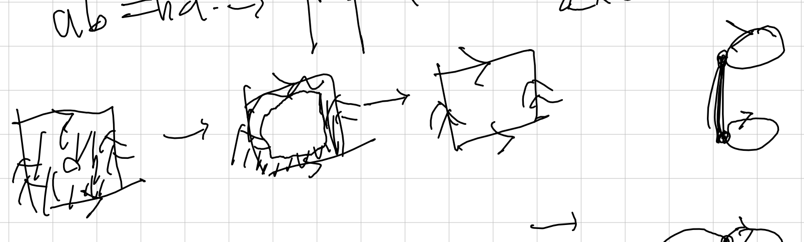
$\langle a, b \mid aba = b \rangle$.

$aba^{-1}b^{-1} = e$.

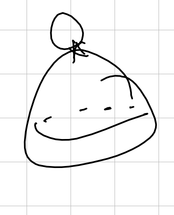
$\langle a, b \mid ab = ba \rangle$
$\mathbb{Z} \times \mathbb{Z}$

$ab = ba \rightarrow$ group $\times \mathbb{Z}$.

$S^1 \vee S^1$



$S^1 \vee S^2$



$T : \mathbb{Z} \times \mathbb{Z}$

$S^1 \vee S^1$
$\mathbb{Z} * \mathbb{Z}$

$T \setminus \circ$ $\mathbb{Z} * \mathbb{Z}$

$K \setminus \circ$ $\mathbb{Z} * \mathbb{Z}$

Theorem

$S^1 \vee S^1$
$\mathbb{Z} \times \mathbb{Z}$

S^1
$\updownarrow$
$\mathbb{Z}$

theorem

$S^3 : 0$.
$\downarrow$
$\mathbb{R}^3 : 0$.

M : n-dim with $n \geq 3$

then $\pi_1(M) \cong \pi_1(M \setminus \circ)$.
$\cong \pi_1(M \setminus 2pt)$
$\cong \pi_1(M \setminus \{x_i\})$

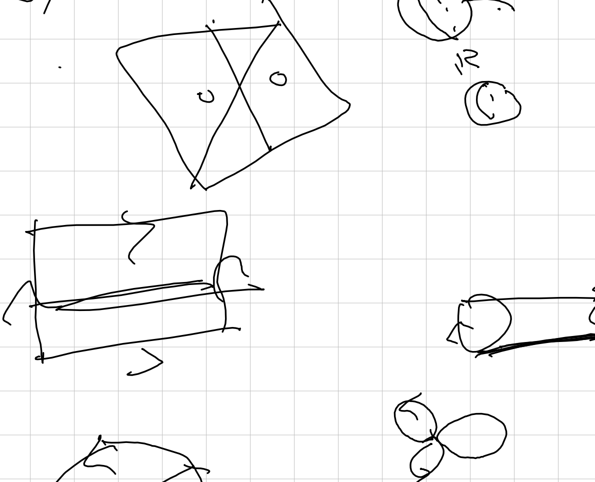
$\pi_1(S^n) = 0$
$n \geq 2$.

$\pi_1(\mathbb{RP}^n) = \mathbb{Z}_2$
$(n \geq 2)$.

tree

$S^1 \vee S^1 \vee \mathbb{R}$.

$\mathbb{Z} * \mathbb{Z}$

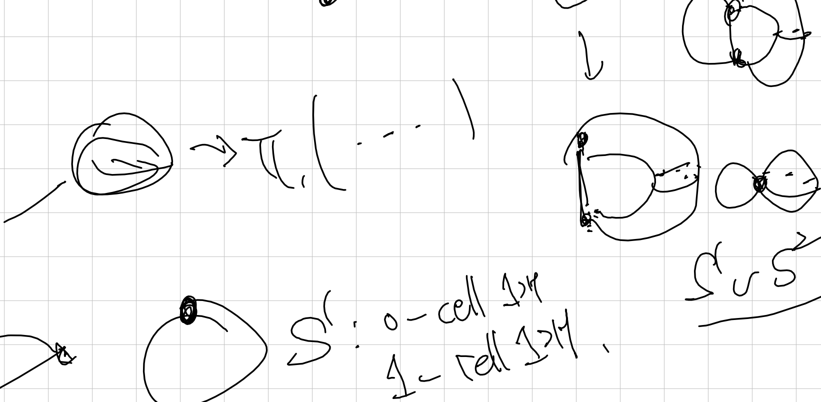


$: S^1 \vee S^1 \vee S^1$

$\mathbb{Z} * \mathbb{Z} * \mathbb{Z}$

Example.

$S^2 / N = S$



$\to \pi_1(\ldots)$

$S^1 \vee S^2$

S^1 : 0-cell 1개
1-cell 1개.

S^2 : 0-cell 1개
2-cell 1개

$S^1 \vee S^1$

$\mathbb{RP}^2$: 0-cell : 1개
1-cell : 1개
2-cell 1개

$aba^{-1}b^{-1} = 0$.

$\langle a, b \mid ab = b \ldots \rangle$

$(\mathbb{Z} * \mathbb{Z}) / a$

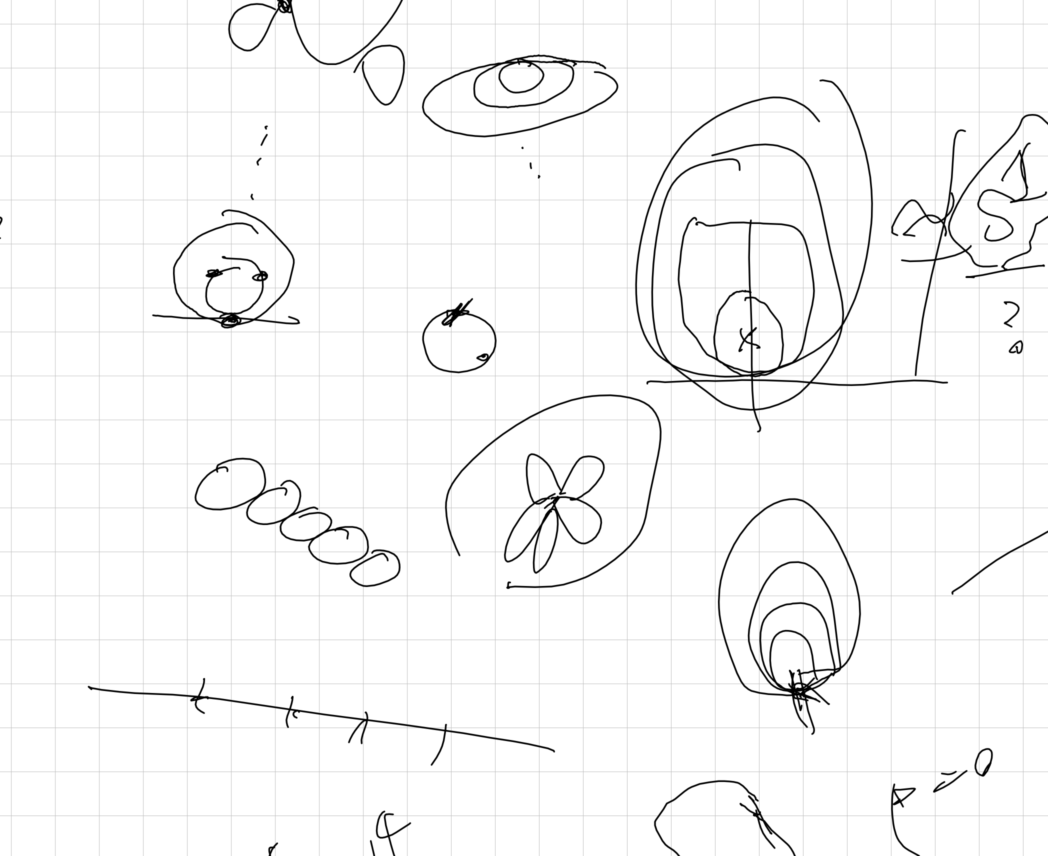
$\mathbb{RP}^2 : \mathbb{Z}_2$ $\langle a \mid a^2 \rangle$
$a^2 = 1$

~~$\mathbb{RP}^3$: 0-cell 1개 1-cell 1개 2-cell 1개 3-cell 1개~~

$\pi_1(\mathbb{RP}^3) = \pi_1(\mathbb{RP}^2) = \mathbb{Z}_2$.

$(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$

$\leftarrow$ $x^2 + y^2 + z^2 = 1$
$\cup$ $x^2 + y^2 \leq 1$
$\cup$ $(0, 0, z)$ $-1 \leq z \leq 1$.



$\mathbb{R}/\mathbb{Z}$

S^1

$z = x^4 + y^4$
$\downarrow$ exp

$K_1 = f(4, 2)$
$K_2 = f_4 \ldots$

$\frac{1}{P_n + z}$.

$\int \frac{R\,dz}{y} \quad \int \frac{1}{z}\,dz$
$\frac{4}{2\pi i}$

$2\pi i \cdot \chi(\ldots)$
$\parallel$
0